



Portfolio

Lab Experiments

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DECLARATION

Roll number F20-BS-PSY-R-1006, student of BS Psychology, session 2020 -2024, hereby declare that the matter presented in the Portfolio is my original work.

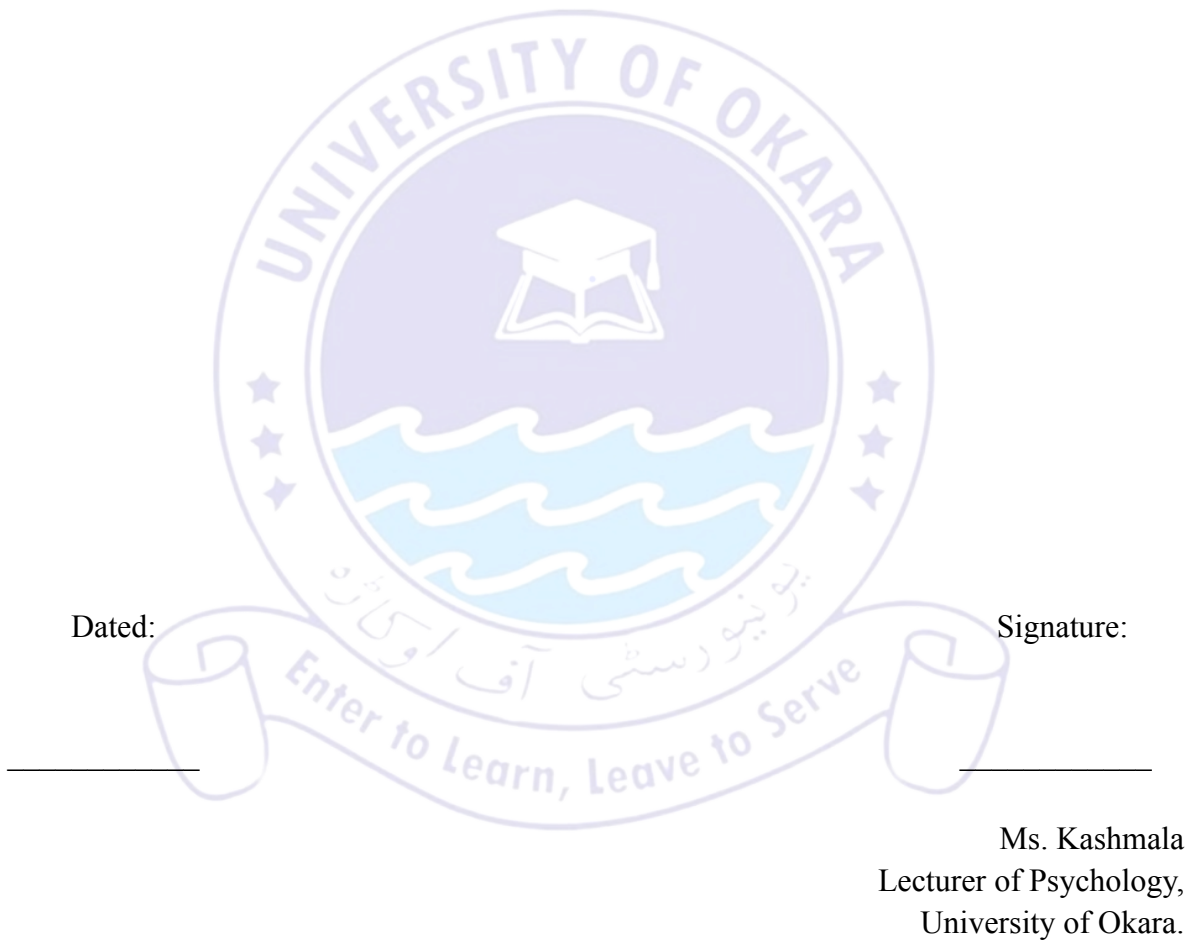
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Signature:



PORTFOLIO COMPLETION CERTIFICATE

It is certified that the work contained in this Portfolio has been completed by F20-BS-PSY-R-1006, a student of BS Psychology, session 2020-2024, under my supervision.



Impact of Operant Conditioning & Children's Behavior

Abstract:

This case study explored the use of operant conditioning, specifically positive reinforcement (continuous schedule of reinforcement), to improve the homework completion timeliness of a 9-year-old girl named Noor. A single-subject design with baseline-intervention-extinction-follow up phases was employed. During the intervention, positive reinforcement (continuous schedule) in the form of extended play time and Lay's snacks were provided for timely homework completion. Results showed a significant decrease in the time taken to complete homework during the intervention compared to the baseline. Extinction and follow-up phases demonstrated the effectiveness of reinforcement and spontaneous recovery of desired behavior, respectively. This study suggests that operant conditioning with positive reinforcement can be an effective tool for parents and educators to promote timely homework completion in children. Further research on reinforcement schedules and long-term maintenance strategies is recommended.

Introduction:

Noor, a bright, and curious 9-year-old girl studying in the 5th grade at a public school in Okara. While Noor possesses a natural thirst for learning, her tendency to procrastinate often leads her to complete her homework late in the evening, causing stress and frustration for both her and her parents. The researcher observed, in the baseline phase, that Noor used to engage in other activities i.e. recurrently going to the washroom, drinking water, or listening to others while studying. She is intelligent but shows non-serious behavior while doing homework.

This research investigates the efficacy of operant conditioning, a theory pioneered by B.F. Skinner, which revolves around the concept that behavior is molded by its outcomes or consequences. It suggests that behaviors are influenced and modified based on the resulting consequences they

produce. Further exploration into this theory aims to deepen our understanding of how behavior is shaped through various reinforcing or punishing consequences. **Operant conditioning** is a method of learning that employs rewards and punishments for behavior. Through operant conditioning, an association is made between a behavior and a consequence (whether negative or positive) for that behavior. **Reinforcement** is any event that strengthens or increases the behavior it follows. There are two kinds of reinforcers. In both of these cases of reinforcement, the behavior increases. **Positive reinforcement** occurs when a certain behavior results in a positive outcome, making the behavior likely to be repeated in the future. This behavioral psychology concept can be used to teach and strengthen behaviors. This study underscores the potential of operant conditioning, employing positive reinforcement, as a valuable tool for parents and educators aiming to encourage punctual homework completion in children. The use of a continuous reinforcement schedule in this study was chosen to establish a robust link between timely homework completion and positive outcomes. **The continuous reinforcement schedule** is one in which the desired behavior is reinforced every single time it occurs. This schedule is best used during the initial stages of learning to create a strong association between the behavior and response. By consistently reinforcing the behavior each time it occurred, it aimed to swiftly strengthen the association between completing homework promptly and receiving positive reinforcement, accelerating the learning process for Noor.

Extinction, within the realm of behavioral conditioning, embodies the fading away of a learned response when the reinforcement ceases. This process entails a gradual decline and eventual cessation of the once-reinforced behavior, illustrating the pivotal role of reinforcement in sustaining desired responses over time. In behavioral psychology, **Extinction** refers to the gradual weakening and eventual disappearance of a conditioned response when it is no longer reinforced.

Spontaneous recovery is a phenomenon in learning and memory, where a previously extinguished conditioned response reappears after a period of rest or decreased responsiveness. In

simpler terms, a behavior previously thought to be gone suddenly returns after a period of inactivity or lessened occurrence.

With Noor, we used a combination of rewards i.e. more play time & favorite snacks and consequences to help her complete her homework on time. By applying the principles of operant conditioning, we aim to make homework a positive and enjoyable experience for her. This will involve setting clear expectations, and providing regular reinforcement, as she develops the habit of timely completion. Noor's motivation will come from positive reinforcement. When she finishes her homework early, she might earn extra playtime and favorite snacks as a reward.

This case study aims to explore how operant conditioning, specifically positive reinforcement, can be used to help Noor develop a more consistent and timely approach to her homework, allowing her to experience the satisfaction of completing her tasks on time and enjoying more play time and her favorite snacks i.e Lay's.

Biodata:

Name : Noor
Age : 9 years
Education : Grade 5
Gender : Female
City : Okara
Siblings : 2 sisters
Birth order : 3rd

Methodology:

This case study will employ a single-subject design with baseline-intervention-extinction-follow up structure. This design will allow for the assessment of Noor's behavior both

before and after the intervention, enabling us to determine the impact of the positive reinforcement (continuous schedule of reinforcement) on her homework completion.

a. Data collection:

1. Daily observations will be conducted to track the timeliness of Noor's homework completion and record her behavior during homework sessions. The checklist will be completed for four weeks.

Data will be collected during the following phases:

- ❖ Week 1: *Baseline* - No intervention implemented.
- ❖ Week 2: *Intervention* - Positive reinforcement applied.
- ❖ Week 3: *Extinction* - Reinforcement withdrawn.
- ❖ Week 4: *Follow up* - Spontaneous Recovery (positive reinforcement provided again).

2. A daily checklist will be used to track the timeliness of homework completion and adherence to the established schedule. The daily checklist will track the following information:

- ❖ **Date:** Date of the observation.
- ❖ **Start Time:** The time homework was started.
- ❖ **Finish Time:** The time homework was completed.
- ❖ **Time Taken:** Time taken to complete homework.
- ❖ **Reinforcement Type:** Positive reinforcement.
- ❖ **Notes:** Any noticeable event should be included.

b. Intervention Procedures:

Clear and consistent expectations regarding the timeliness of the homework completion schedule will be communicated to Noor and her parents. Reinforcement will be applied consistently

throughout the intervention period to maintain their effectiveness. Positive reinforcement (continuous schedule of reinforcement) was used to modify the behavior.

Assessment:

To thoroughly assess the effectiveness of interventions, a well-rounded strategy using both formal and informal evaluation methods is crucial. This case study investigated the impact of positive reinforcement on Noor's timeliness of homework completion by employing a combination of formal and informal assessment methods, providing a comprehensive understanding of her behavioral changes.

Formal Assessment:

A daily checklist was used to track Noor's timeliness of homework completion habit. The checklist recorded the day, start time, finish time, time taken, and reinforcement provided.

Weekly Assessment of Homework Completion

<i>Phase</i>	<i>Start Time (Avg.)</i>	<i>Finish Time (Avg.)</i>	<i>Time Taken (finish time - start time)</i>	<i>Reinforcement Provided</i>
Baseline	3 PM	9 PM	6 hours	N/A
Intervention	3 PM	7 PM	4 hours	Positive (playtime and Lay's)
Extinction	3 PM	8 PM	5 hours	N/A
Follow-up	3 PM	7 PM	4 hours	Positive (playtime and Lay's)

Baseline - Week 1

<i>Day</i>	<i>Start Time (Avg.)</i>	<i>Finish Time (Avg.)</i>	<i>TimeTaken (finish time - start time)</i>	<i>Reinforcement Provided</i>
Monday	3 PM	9 PM	6 hours	N/A
Tuesday	3 PM	9 PM	6 hours	N/A
Wednesday	3 PM	9 PM	6 hours	N/A
Thursday	3 PM	9 PM	6 hours	N/A
Friday	3 PM	9 PM	6 hours	N/A
Saturday	3 PM	9 PM	6 hours	N/A
Sunday	3 PM	9 PM	6 hours	N/A

Intervention - Week 2

<i>Day</i>	<i>Start Time (Avg.)</i>	<i>Finish Time (Avg.)</i>	<i>TimeTaken (finish time - start time)</i>	<i>Reinforcement Provided</i>
Monday	3 PM	7 PM	4 hours	Positive (playtime and Lay's)
Tuesday	3 PM	7 PM	4 hours	Positive (playtime and Lay's)
Wednesday	3 PM	7 PM	4 hours	Positive (playtime and Lay's)
Thursday	3 PM	7 PM	4 hours	Positive (playtime and Lay's)
Friday	3 PM	7 PM	4 hours	Positive (playtime and Lay's)
Saturday	3 PM	7 PM	4 hours	Positive (playtime and Lay's)
Sunday	3 PM	7 PM	4 hours	Positive (playtime and Lay's)

Extinction - Week 3

<i>Day</i>	<i>Start Time (Avg.)</i>	<i>Finish Time (Avg.)</i>	<i>TimeTaken (finish time - start time)</i>	<i>Reinforcement Provided</i>
Monday	3 PM	7 PM	4 hours	N/A
Tuesday	3 PM	7 PM	4 hours	N/A
Wednesday	3 PM	9 PM	6 hours	N/A
Thursday	3 PM	9 PM	6 hours	N/A
Friday	3 PM	9 PM	6 hours	N/A
Saturday	3 PM	9 PM	6 hours	N/A
Sunday	3 PM	9 PM	6 hours	N/A

Follow up - Week 4

<i>Day</i>	<i>Start Time (Avg.)</i>	<i>Finish Time (Avg.)</i>	<i>TimeTaken (finish time - start time)</i>	<i>Reinforcement Provided</i>
Monday	3 PM	9 PM	6 hours	Positive (playtime and Lay's)
Tuesday	3 PM	7 PM	4 hours	Positive (playtime and Lay's)
Wednesday	3 PM	7 PM	4 hours	Positive (playtime and Lay's)
Thursday	3 PM	7 PM	4 hours	Positive (playtime and Lay's)
Friday	3 PM	7 PM	4 hours	Positive (playtime and Lay's)
Saturday	3 PM	7 PM	4 hours	Positive (playtime and Lay's)
Sunday	3 PM	7 PM	4 hours	Positive (playtime and Lay's)

During the **baseline phase**, the child consistently completed homework late, averaging 6 hours. The **intervention phase** saw the introduction of consistent positive reinforcement (continuous schedule of reinforcement), which led to the achievement of target behavior. The average timeliness of the homework completion dropped to 4 hours. The extinction of reinforcement during the **extinction phase** dropped her behavior back to the baseline phase after two days. Finally, the **follow-up phase** showed spontaneous recovery, due to positive reinforcement.

Informal Assessment:

Beyond formal assessment methods, informal observations played a crucial role in evaluating the intervention's effectiveness. Daily observations of Noor during homework sessions provided valuable insights into her engagement, focus, and overall attitude toward the task. Noor exhibited increased interest in completing her homework on time due to reinforcement, but she couldn't maintain this routine during the last five days of the extinction phase because the reinforcers were removed. Reinforcers also aided her in staying focused on her homework despite distractions. Additionally, she became less engaged in other activities, such as frequent bathroom breaks, drinking water, or being distracted by others while studying, and became more serious during her study hours

in the intervention phase. Moreover, she displayed improved politeness and attentiveness to instructions. She willingly approached her diary, completing homework promptly to receive reinforcement, effectively overcoming procrastination.

Pre-Assessment:

Before implementing the positive reinforcement intervention, Noor's homework completion was assessed during a week-long baseline phase. Daily observations tracked her start time, finish time, total time taken, and any noticeable behaviors. This phase revealed a consistent pattern of late completion, averaging 6 hours for homework tasks. Furthermore, During homework sessions, Noor frequently engaged in non-serious behaviors like going to the washroom, drinking water, and listening to others, indicating a lack of focus and engagement with the task. The pre-assessment revealed:

- ❖ Average completion time: 6 hours, indicating consistent late completion.
- ❖ Observations: Noor exhibited distractions, low engagement, and frustration during homework sessions.

Post-Assessment:

Following the intervention phase, which introduced consistent positive reinforcement (playtime and Lay's) for timely homework completion, Noor's behavior demonstrated significant improvement. The average completion time dropped to 4 hours, showcasing a reduction of 2 hours. This positive change suggests that the reinforcement effectively shaped Noor's behavior, encouraging her to focus and complete her homework more efficiently. Additionally, informal observations during homework sessions revealed increased engagement, focus, and a more positive attitude towards the task, further supporting the effectiveness of the intervention.

The extinction phase, where reinforcement was withdrawn, confirmed the dependence on the external reward system. Noor's behavior regressed to the baseline level, completing homework within an average of 5 hours.

Finally, the follow-up phase, where positive reinforcement was reintroduced, demonstrated spontaneous recovery. Noor's average completion time returned to 4 hours, suggesting that the initial learning had not completely extinguished. This finding implies that the positive reinforcement had a lasting impact on her behavior, making her more receptive to resuming the desired behavior when the reward was reintroduced.

Conclusion:

This case study provides compelling evidence for the effectiveness of operant conditioning in shaping children's behavior. By employing positive reinforcement, coupled with clear expectations and consistent application, the intervention successfully instilled the desired habit of timely homework completion in the subject. The removal of reinforcement (extinction) caused a decline in learned behavior and the reintroduction of reinforcement (spontaneous recovery) proved crucial in maintaining positive behavior change over time.

This study suggests that operant conditioning can be a powerful tool for parents and educators in fostering positive and productive behaviors in children. Future research delving into the specifics of reinforcement schedules, behavior types, and long-term maintenance strategies will further strengthen the application of this approach in promoting desirable behaviors among diverse populations and settings.